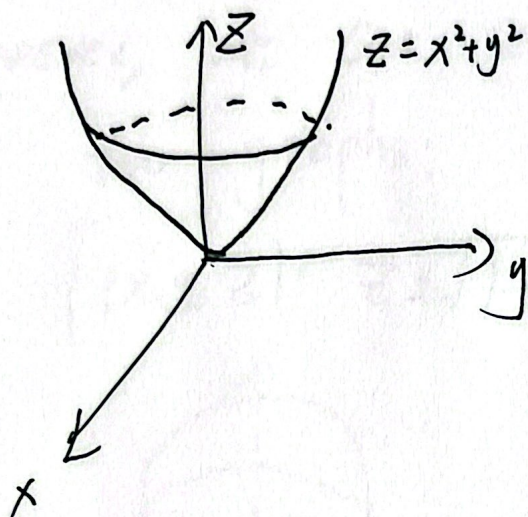


Sep 14.

Graph of a function

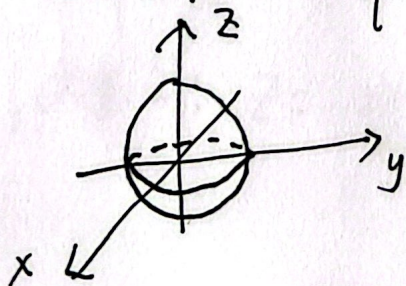
Given a function ~~So~~ $f(x,y) = x^2 + y^2$, we consider its graph $\{(x,y,z) : z = f(x,y)\}$.



Question: Can a vertical line parallel to the (z) -axis intersect this graph twice?

No. the function $z = f(x,y)$ uniquely determine the z coordinate of a point ~~whose~~ (x,y) coordinate.

Now consider a sphere $x^2 + y^2 + z^2 = 1$.



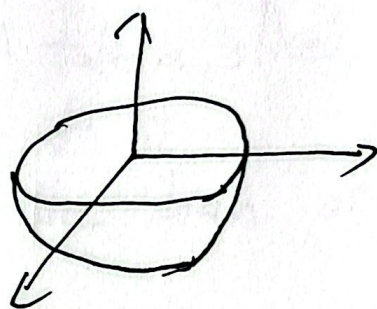
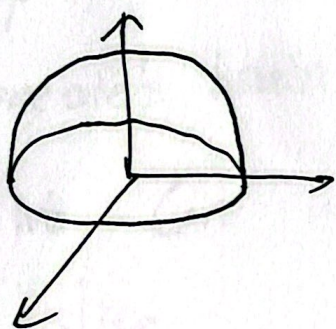
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Question: Can you see this sphere as a graph of a function?

No. the z -axis intersect the sphere at two points $(0,0,1)$, $(0,0,-1)$

However, we can see this sphere as union of two graphs:

$$\{(x,y,z): z = \sqrt{1-x^2-y^2}\} \quad \{(x,y,z): z = -\sqrt{1-x^2-y^2}\}$$



Transformation of geometric subjects.

① Translations:

$$\{(x,y,z): z = x^2 + y^2\} \longrightarrow \{(x,y,z): z = (x-1)^2 + (y-2)^2\}$$

$$\{(x,y,z): z = x+y\} \longrightarrow \{(x,y,z): z = x+y+1\}$$

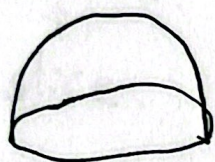
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②. Reflection.

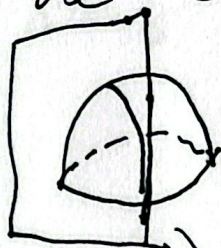
$$\{(x, y, z) : z = \sqrt{1 - x^2 - y^2}\} \rightarrow \{(x, y, z) : z = -\sqrt{1 - x^2 - y^2}\}$$

Using cross-sections to reveal the shape.

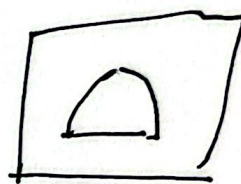
Recall that how to use CT in hospital.



↑
Someone's brain



take a slice
of it.

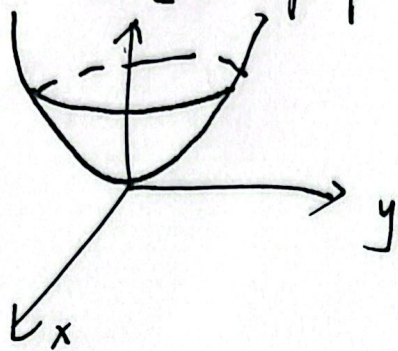


Then we can understand the brain at least at this slice.

The more slices we take, the more information we get.

We can do the same thing for the graph of a function!

Consider the z graph of $z = x^2 + y^2$.



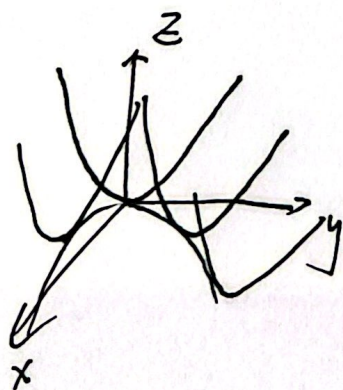
Consider the intersection of this graph with $\{y=a\}$. Here " a " is a number.
 $\uparrow$
 a slice.

When we move a , we get different slices.

$$\{(x,y,z) : z = x^2 + y^2\} \cap \{(x,y,z) : y = a\} = \{(x,y,z) : z = x^2 + a^2\}$$

This is a ~~prg~~ paraboloid

Exercise: get slices for the graph of $z = x^2 - y^2$.



sep 14 (2)